MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. The value of $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta}$ is equal to: [1]

(A) $2\sec^2\theta$ (B) $2\cos^2\theta$
(C) $2\sin^2\theta$ (D) $\sec^2\theta$

Q2. If $\csc\theta - \cot\theta = \frac{1}{3}$, then the value of $\csc\theta + \cot\theta$ is: [1]

(A) $\frac{1}{3}$ (B) 3 (C) $\frac{2}{3}$ (D) 1

Q3. The expression $(\sin^4\theta - \cos^4\theta + 1)\csc^2\theta$ simplifies to: [1]

(A) 1 (B) 2 (C) 0 (D) -1

Q4. If $k + 1 = \sec^2\theta(1 + \sin\theta)(1 - \sin\theta)$, then the value of k is: [1]

(A) 0 (B) 1 (C) 2 (D) -1

Q5. The value of $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}}$ for an acute angle θ is equal to: [1]

(A) $\csc\theta + \cot\theta$
(B) $\sec\theta + \tan\theta$
(C) $\csc\theta - \cot\theta$
(D) $\sec\theta - \tan\theta$

Q6. **Assertion (A):** In a right-angled triangle ABC with $\angle B = 90°$, $\sin A = \cos C$.
Reason (R): $\sin A = \frac{BC}{AC}$ and $\cos C = \frac{BC}{AC}$, so both are identical side ratios. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Prove the identity: $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. [2]

Q8. Express $\sin A$, $\sec A$, and $\tan A$ in terms of $\cot A$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Prove that: $(1 + \tan A \tan B)^2 + (\tan A - \tan B)^2 = \sec^2 A \sec^2 B$. [3]

Q10. If $\sin\theta = \frac{c}{\sqrt{c^2+d^2}}$ $(d > 0)$, find the values of $\cos\theta$ and $\tan\theta$. [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

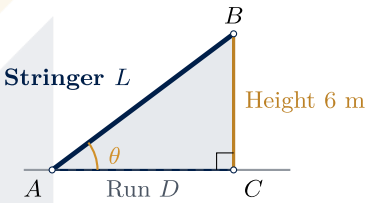
Q11. If $\sec\theta = x + \frac{1}{4x}$, prove that $\sec\theta + \tan\theta = 2x$ or $\frac{1}{2x}$. [5]

Q12. If $\sin\theta + \cos\theta = p$ and $\sec\theta + \csc\theta = q$, prove that $q(p^2 - 1) = 2p$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Pedestrian Foot Overbridge Staircase Design

An urban transit planner designs a straight public staircase for a foot overbridge. The total vertical height to be climbed is 6 m. Each step has a riser (height) of 15 cm and a tread (depth) of 20 cm, establishing an inclination angle θ for the stringer beam such that $\tan\theta = \frac{15}{20} = \frac{3}{4}$.



Based on the above geometric model, answer the following questions:

(i) Determine the values of $\sin\theta$ and $\cos\theta$ for the staircase stringer incline. [1]

(ii) Calculate the total horizontal run D of the staircase. [1]

(iii) Calculate the exact length L of the diagonal handrail beam spanning from the bottom landing to the bridge platform. [2]

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